

RBF-PSO-IS: An innovative metamodeling for reliability analysis of bridge's vortex-induced vibration

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ABSTRACT

The complexity of vortex-induced vibration (VIV) instability analysis originates from the variability of the wind environment, the randomness of the analytical model, the ambiguity of the failure criterion, and the high-dimensional nonlinearity and non-explicitness nature of the instability function. This paper addresses the low sampling efficiency and challenges in approximating failure probability in prior research on reliability. By integrating an adaptive sampling strategy with particle swarm optimization (PSO), an initial VIV reliability metamodel is established, which is subsequently resampled to create a more refined metamodel. A combination of multiple reliability algorithms is then employed to calculate the maximum amplitude of VIV and the failure probability of the VIV interval. The validity and accuracy of the proposed method are demonstrated through two representative numerical examples. The results indicate that the proposed radial basis function (RBF)-PSO-importance sampling (IS) analysis method and its adaptive random sampling strategy can significantly reduce the computational cost associated with extensive finite element model operations and the complexity arising from the high-dimensional nonlinearity and non-significance of the limit state function (LSF). This method enables researchers to achieve more accurate and time-saving solutions for VIV in steel-concrete composite bridges.

1. Introduction

Steel-concrete composite beams are widely employed in suspension bridges due to their open cross-sections, which makes them susceptible to periodic vortex shedding under wind loads. Their poor torsional stiffness and lightweight nature render them prone to significant VIV. Numerous studies have investigated the VIV of steel-concrete composite beams. Ge and Xiang [1] evaluated the VIV occurrence probability of suspension bridges using the Scanlan vortex shedding semi-empirical model. Li et al. [2] assessed the reliability of VIV in large-span suspension bridges, considering uncertainties in wind flow fields, structural properties, and wind-bridge interactions. Zhou et al. [3] employed the Multiplicative Dimensional Reduction Method for VIV reliability assessment, accounting for wind and structural uncertainties.

The stability of bridges (VIV) is a reliability issue involving high-dimensional nonlinear and implicit functional functions, which are challenging to solve with satisfactory accuracy using traditional first-order, second-moment, and response surface methods [4]. Fang et al. [5] and Hu et al. [6] employed the Monte Carlo technique to analyze the flutter reliability of long-span bridges under the influence of tropical

cyclone winds and to quantify the uncertainty caused by limited samples in quantile estimation. Although direct sampling (DS) Monte Carlo simulations (MCS) offer high confidence, they are computationally intensive. Researchers often employ variance reduction techniques and metamodel techniques to reduce the computational cost of Monte Carlo methods [7]. Variance reduction techniques accelerate convergence by reducing sample variance but still require tens of thousands of samples. Metamodels use low-complexity mathematical models, such as artificial neural networks [7], Kriging models [8], and support vector machines (SVM) [9], to approximate the mapping relationship between random variables and responses, thus reducing the computation time of a single sample. Most previous reliability studies [10–14] relied solely on metamodels for reliability analysis, with accuracy rarely approaching the analytical solution.

To address these issues, this paper combines the advantages of variance reduction technology and metamodels. Initially, a minimum number of samples are retained for constructing the metamodel to evaluate the LSF. Subsequently, variance reduction technology is applied to the metamodel, obtaining a more accurate mapping relationship. Among various metamodels, the RBF neural network is known

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for its high learning efficiency, nonlinear mapping ability, and absence of local minimum issues. Consequently, this paper employs the RBF neural network to build the metamodel.

In this study, the reliability of VIV is evaluated using a large-span suspension bridge as the subject. The research accounts for uncertainties related to wind attack angle, wind direction, and structural damping, and addresses the ambiguity in damage criteria, which has not been considered in previous VIV studies [2,3,15]. This paper introduces RBF-PSO-IS method for VIV analysis. Employing the RBF neural network and an adaptive random sampling strategy, a metamodel is constructed and optimized using the PSO algorithm. By adopting the IS method for sampling the metamodel, the sampling center of gravity shifts towards the vicinity of the most probable failure point. This approach obtains samples with larger contributions and allows for the reconstruction of the metamodel. Furthermore, the reconstructed metamodel is combined with various reliability algorithms to calculate the maximum amplitude of VIV and the failure probability within VIV intervals.

In summary, this paper presents a comprehensive approach to VIV reliability assessment in large-span suspension bridges by addressing uncertainties in wind attack angle, wind direction, structural damping, and damage criteria. The proposed RBF-PSO-IS method is expected to contribute significantly to the field of structural engineering and provide valuable insights for the design and analysis of suspension bridges under VIV.

2. Theoretical basis of failure probability analysis

2.1. VIV parameter uncertainty analysis

The decision tree model can be employed for the probabilistic prediction of sporadic VIV events measured by the bridge monitoring system. Uncertainties associated with structural and wind properties are taken into account (Fig. 1). Consequently, the likelihood of a bridge VIV occurring is determined, allowing the management department to implement appropriate and timely countermeasures. The prediction of bridge VIV considers uncertainties associated with damping ratio, wind speed, wind direction, and wind angle of attack as random variables. For VIV to occur, the following conditions must be met simultaneously: the damping ratio is lower than a certain threshold; the wind direction corresponds to the direction of VIV; the wind angle corresponds to the angle of VIV; and the wind speed falls within the lock-in wind speed interval of VIV. These conditions for VIV occurrence can be expressed as branch 1 in Fig. 1. If any of the aforementioned conditions are not satisfied, VIV will not occur.

In addition to the randomness of damping, wind angle of attack [3], wind direction, and wind speed, this study also considers the ambiguity in the failure criterion, reflecting the uncertainty in the VIV response and probability of failure.

2.2. VIV amplitude

VIV occurs when the maximum VIV amplitude exceeds the specified allowable limit. The VIV amplitude evaluation method (JTG/T3360-01–2018 [16]) is too simplified and suitable for bridges with a span of less than 200 m. Therefore, it is necessary to establish an improved VIV performance function, which is applied to large-span bridges. VIV is a kind of vibration with one degree of freedom, so the reaction of kinetic energy and impulse, represented by $(2\pi)^5 z_0^3 f_h^5$, is used as the measurement criteria to evaluate vehicle driving stability. The Sperling index is employed to assess driving comfort, as listed in Table 1. The empirical equation of vehicle driving stability [17] is established as follows:

$$W = 2.7 \sqrt[10]{z_0^3 f_h^5 F(f_h)} = 0.896 \sqrt[10]{a^3 F(f_h) f_h} \quad (1)$$

where, z_0, f_h and $a = z_0 2\pi f_h^2$ represent amplitude (cm), vertical vibration frequency (Hz), and acceleration ($\text{cm} \cdot \text{s}^{-2}$), respectively. $F(f_h)$ is the weighting factor of vertical vibration. $F(f_h) = 0.325 f_h^2$ when $f_h = 0.5$ – 5.9 Hz. W is the basic requirement for driving comfort; the amplitude required to achieve driving comfort can be determined by substituting W and frequency f_h into Eq.(6). Driving comfort can then be assessed according to Table 1.

If the maximum VIV amplitude exceeds the allowed value, the VIV amplitude is considered to be in a failure state. In such cases, appropriate countermeasures should be taken to control VIV. The performance function of the bridge VIV amplitude can be obtained through LSF:

$$G(y_{\max}, [y_a]) = [y_a] - y_{\max} \quad (2)$$

where, $y_{\max}$ is the maximum VIV amplitude of the bridge, and $[y_a]$ represents the z_0 aorth noting that this research method does not make specific assumptions about the nature of structural LSF and random variables. Therefore, by adjusting the numerical mods.

3. Efficient failure probability algorithm

3.1. Efficient algorithm of failure probability

In this paper, the generalized Monte Carlo method is approximated,

Table 1
Sperling evaluation criteria.

W	Ride comfort (feeling for vibration)
2	Clearly feel
2.5	More obvious, but not unpleasant
3	Intense, abnormal, but tolerable
3.25	Very abnormal
3.5	Extremely abnormal, disgusting, annoying, unbearable for a long time
4	Extremely disgusting and harmful to bear for a long time

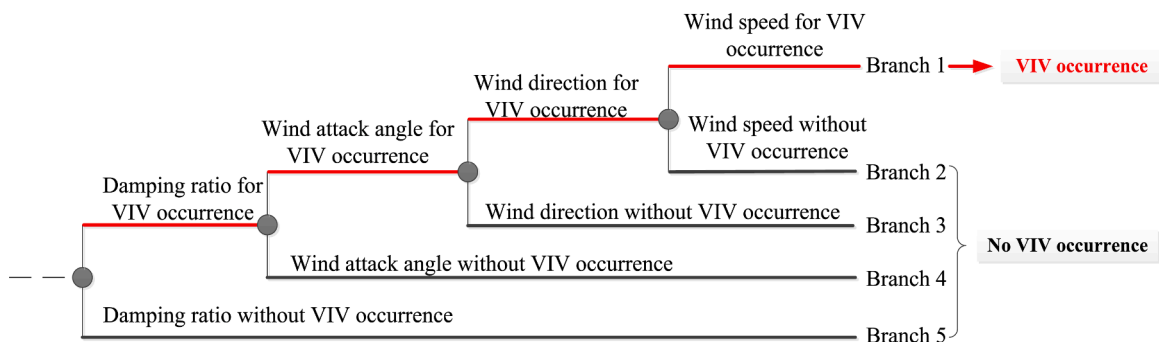


Fig. 1. Decision tree for predicting VIV occurrence.

i.e., the probability of failure is estimated by the frequency of failure samples occurring in the random sampling process, $\hat{P}_f$. The MCS generates a large number of sample points and verifies whether each sample point is located in the failure region. The structural damage probability is defined as:

$$\hat{P}_f = \frac{1}{N_{MCS}} \sum_{i=1}^{N_{MCS}} I[G(x_i)] \quad (3)$$

where, $I(x)$ is a failure indicator function. It is specified that when $x < 0$, $I(x) = 1$; when $x \geq 0$, $I(x) = 0$. $G(x_i)$ is the LSF. The LSF $G(x_i) = 0$ divides the design space into the failure domain $G(x_i) \leq 0$ and the safety domain $G(x_i) > 0$. N_{MCS} is the number of random samples. With the increase of N_{MCS} , $\hat{P}_f$ gradually converges to the true solution with the coefficient of variation:

$$\delta_{MCS} = \sqrt{\frac{1 - \hat{P}_f}{\hat{P}_f N_{MCS}}} \quad (4)$$

3.2. RBF neural network

In constructing an RBF neural network, it is necessary to have a training set $X_i (i = 1, 2, \dots, m)$ and a response set $Y_i (i = 1, 2, \dots, n)$. RBF neural networks are feed-forward networks capable of approximating any continuous function with arbitrary accuracy. During the training and application of RBF, it is essential to compare the true value with the predicted value and adjust the link weight of RBF according to the error backpropagation using feedback, ultimately completing the learning process. The structure of the RBF neural network is shown in Fig. 2.

The trained RBF model can serve as an alternative to the time-consuming finite metamodel in structural analysis, thereby enhancing efficiency and reducing computational costs. The results are used to construct a metamodel, which is typically several orders of magnitude faster than actual simulations, leading to a significant reduction in computation time.

In RBF neural networks, crucial parameters for function fitting include the distribution density of the radial basis function and the number of neurons. As the sample size increases, these parameters should be re-determined. Consequently, the RBF neural network is

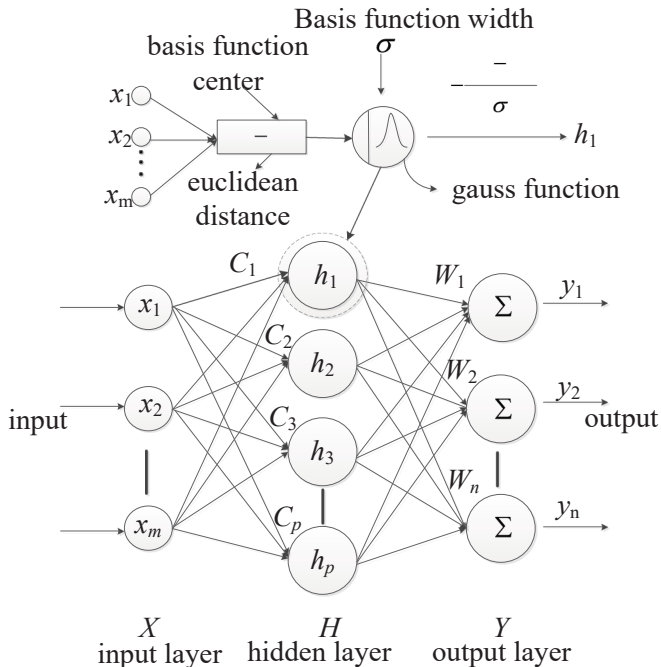


Fig. 2. RBF network structure.

improved by various evolutionary algorithms [18] that continuously adjust parameters within the neural network, thereby bringing the model's performance closer to the dataset's.

3.3. Importance sampling

The direct sampling method often results in sample points primarily concentrated near the maximum point of the probability density function, with few sample points falling within the failure domain. Consequently, instances of $G(u) < 0$ are rare. To address the limitations of MCS, researchers have proposed various variance reduction techniques that aim to decrease the total number of performance function calls needed to accurately estimate failure probability. These techniques include IS and subset simulation (SS) methods [19]. IS is considered one of the most effective variance reduction approaches. By shifting the center of random sampling, sample points are more likely to fall within the failure domain, thereby increasing the probability of $G(u) < 0$ (as shown in Fig. 3). This concept forms the basis of the IS method for MCS of structural reliability. In the standard normal space, the expression for structural failure probability can be written as:

$$\hat{P}_f = \int_{\mathbb{R}^n} \frac{I[G(u)]f(u)}{\varphi_n} \varphi_n(u) du = E \left\{ \frac{I[G(u)]f(u)}{\varphi_n} \right\} \quad (5)$$

where, $f(u)$ is the probability density function, and φ_n represents the probability density function used for sampling, with the normal distribution probability density employed.

This paper suggests that the sampling center can be placed at point P^* within the failure domain, contributing the most to the probability of structural failure, i.e., the most probable failure point [20]. This point can be obtained by solving the following constrained optimization problem:

$$\begin{cases} \min_u \|u\| \\ \text{s.t. } \hat{G}(u) = 0 \end{cases} \quad (6)$$

As a result, a specific iterative calculation formula [21] is established, which addresses the non-convergence issue while reducing operational costs.

3.4. Adaptive random sampling method

Before using a metamodel to solve the failure probability, it is necessary to obtain sample points in advance to train the model to meet certain accuracy requirements. Consequently, the experimental design method of training samples becomes a crucial factor in determining the prediction accuracy of the metamodel. Sampling process-based experimental design methods can be classified as direct sampling methods and sequential sampling methods [4]. As evident in Eq. (14), the magnitude of the predicted value of the metamodel in the reliability problem is not important, while its positive or negative value is the focus. In other words, the metamodel only needs a sufficiently accurate approximation

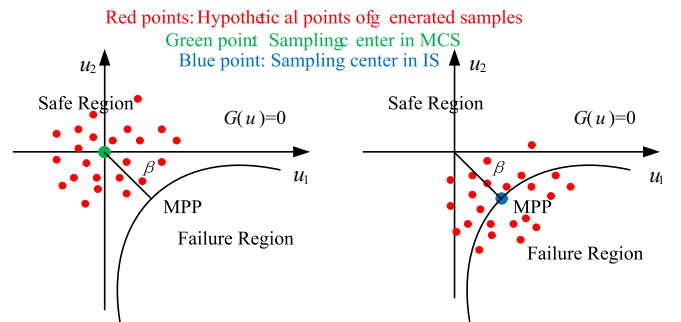


Fig. 3. The process of adding important sampling points.

of the LSF near the LSF, especially near the most likely failure point. Thus, high-value sample points in the reliability analysis should possess the following characteristics [10]:

- Close to the LSF, i.e., $|G(X)| \approx 0$.
- The failure mode is more likely to occur, i.e., the sample point probability density function value $f(X)$ is larger.

In light of this, this paper proposes an adaptive strategy based on weighted sampling and dynamic distance constraints. First, the samples $x_i (i = 1, 2, \dots, N_{\text{MCS}})$ generated by the MCS are converted into standard normal space (U-space) samples $u_i (i = 1, 2, \dots, N_{\text{MCS}})$ using Rosenblatt transform or Nataf [22] transform. The corresponding LSF values are predicted using the current metamodel. Then, the first m sample points closest to the LSF that meet the constraints of Eq. (7) are selected.

$$\begin{cases} d_{u^*} \geq D \\ d_{u^*} = \min\{ \|u^* - u_i\| \}, i = 1, 2, \dots, N_{\text{doe}} \end{cases} \quad (7)$$

where, d_{u^*} is the minimum Euclidean distance between sample u^* and the existing N_{doe} training samples, and D is the distance threshold, which can be half of the maximum of the minimum Euclidean distance between the existing training samples [23], i.e.

$$D = \max\{D_i | D_i = 0.5 \min\{\|X_i - X_j\|, i \neq j, i, j = 1, 2, \dots, N_{\text{doe}}\}\} \quad (8)$$

Finally, weighted sampling is performed to generate m random numbers r_i in the interval $[0.5, 1]$. The importance coefficients of each sample are calculated according to the following equation:

$$c_i = \log r_i \times \|u_i\| \quad (9)$$

The m samples are arranged in descending order by c_i values, and the first q samples are selected as new training samples. Usually, m can be 5, 10, or 15, and q can be 1 or 2. The distance constraint ensures that the selected sample points are evenly distributed around the LSF, avoiding the phenomenon of “clustering” of sample points and thus reducing redundant information [24]. The weighted sampling is used to select more failure sample points with higher probability density values and to consider the sample points at the edges of the design space.

3.5. Process of RBF-PSO-IS method

The reliability analysis process using the RBF-PSO-IS method is presented, primarily focusing on adding the “potential” P^* to the metamodel for refinement, implementing the MCS method to predict the failure probability, and iterating these steps until the failure probability converges and the metamodel is preliminarily determined. Subsequently, the IS method is employed to resample the metamodel, which is reconstructed to predict the failure probability.

In summary, the specific steps of the RBF-PSO-IS method are as follows:

(1) Generate the initial training set. M sample points are created in the standard normal space using the Latin hypercube sampling (LHS) method to generate the initial training set. Subsequently, the k-means clustering method [25] is employed to select N_{doe} sample points from the M sample points. These sample points are then transformed into corresponding non-Gaussian variables $x_i (i = 1, 2, \dots, N_{\text{doe}})$ using either the Rosenblatt transform or the Nataf transform and substituted into the finite metamodel to obtain the initial training set:

$$S_{\text{doe}} = \{(u_i, G(x_i)) | i = 1, 2, \dots, N_{\text{doe}}\} \quad (10)$$

The number of initial training sets can be estimated by Liu et al. [24]:

$$N_{\text{doe}} = \max\{2N + 1, 0.5N_{\text{max}}\} \quad (11)$$

where, N is the dimension of random variables, and N_{max} allows the number of calls for finite element analysis.

(2) Generate MCS samples. The N_{MCS} sample points are generated using **step (1)** sampling method in the standard normal space. Currently, these sample points do not need to call the finite metamodel.

(3) Create the RBF model. Based on the initial training set S_{doe} , evolutionary algorithms are used to optimize the parameters of the RBF neural network. At the same time, the leave-one-out method [26,27] is employed to evaluate model performance and prevent overfitting. The MATLAB Neural Network toolbox is used here.

(4) Update the training set. The MCS samples are substituted into the RBF model for solving, and q new training sample points u^* are selected according to the adaptive sampling strategy defined in Section 3.4, substituted into the true LSF for solving, obtaining $G(x^*)$, and adding the new sample $S_{\text{add}} = \{(u^*, G(x^*))\}$ to the updated training set $S_{\text{doe}} = S_{\text{doe}} \cup S_{\text{add}}$. The training set is continuously updated S_{doe} , and a new RBF model is built.

(5) Calculate the probability of failure. Based on the retrained RBF model and MCS samples, the probability failure $\hat{P}_f$ is calculated using Eq. (14).

(6) Determine the convergence criterion. During the iterative process, if the relative change rate of the calculated failure probability $|\hat{P}_{f_{\text{new}}} - \hat{P}_{f_{\text{old}}}| / \hat{P}_{f_{\text{old}}} < \varepsilon$ is less than ε (typically ε less than 0.005), and the number of added new training sample points is greater than the minimum allowable amount, proceed to the next step; otherwise, repeat steps 4–5.

(7) Calculate the coefficient of variation. According to Eq. (4), the coefficient of variation of $\hat{P}_f$ is calculated. If the coefficient of variation is less than 0.05, the process proceeds to the next step. Otherwise, return to **step 3** and generate more sample points, retaining the previously constructed RBF neural network information. Then go to **step 4** and repeat the above process until all convergence requirements are met and the metamodel is initially determined.

(8) Resampling. Based on the established metamodel, the most probable failure point P^* is identified using the method described in Section 3.3. Subsequently, P^* is used as the centroid for resampling, and the normal distribution probability density function is employed as the sampling function for the resampling process.

(9) Updating the training set. Initially, the resampled samples u are input into the metamodel to compute $\hat{G}(x)$. Subsequently, based on the absolute $\hat{G}(x)$ values, the sample points u are uniformly divided into m segments. A sample point u is then randomly chosen from each segment as a new training sample and substituted into the true LSF for solving, yielding $G(x^*)$. Ultimately, the new samples $S_{\text{add}} = \{(u^*, G(x^*))\}$ are integrated into the updated training set $S_{\text{doe}} = S_{\text{doe}} \cup S_{\text{add}}$. The training set is continuously updated S_{doe} , and a new RBF model is built.

(10) Calculate the failure probability. Calculate the failure probability $\hat{P}_f$ using Eq. (5) based on the retrained RBF model and IS.

(11) Determination of convergence criteria. Suppose the relative change rate of the failure probability calculated twice during the iterative process is less than the tolerance range (usually 0.005), and the number of new training sample points added is greater than the minimum allowed number. In that case, the process is considered converged, and the final failure probability can be output. If these conditions are not met, repeat steps 9–10 until convergence is achieved.

In summary, the constructed metamodel is referred to as RBF-PSO-IS. This paper utilizes the developed RBF-PSO-IS method in conjunction with various reliability analysis techniques to study failure probabilities. The reliability methods include the Rackwitz-Fiessler (JC) method, Breitung method, and different Monte Carlo techniques such as direct sampling (DS) method, IS method, and LHS method. The accuracy of the metamodel is verified by comparing the VIV failure probabilities obtained from the original LSF and the metamodel, both using the same reliability method. The aforementioned reliability method is detailed by Zhang and Jin [21].

The main steps of RBF-PSO-IS are divided into two stages. Stage RBF-

PSO: **steps 1–7**, preliminarily determining the metamodel. Stage RBF-PSO-IS: **steps 1–11**, ultimately determining the metamodel. The example demonstrates the effectiveness of the RBF-PSO-IS algorithm. First, we test a 2D example with a low failure probability, which facilitates the demonstration of adding sample points. Then, the method is applied to high-dimensional nonlinear bridge VIV cases, verifying the feasibility of RBF-PSO-IS in bridge VIV analysis.

4. A 2D analysis example

In this example, a two-dimensional random variable model with a small failure probability is employed, and the LSF is represented as:

$$H(U_1, U_2) = 0.5(U_1 - 2)^2 - 1.5(U_2 - 5)^3 - 3 \quad (12)$$

where, the two independent random variables, U_1 and U_2 , are standard and normally distributed.

In order to analyze the accuracy and efficiency of the preliminary RBF-PSO metamodel and the influence of the initial sample point quantity in the RBF-PSO stage, the LHS method was first employed to generate 5×10^7 points. These points were input into the limit state function to establish a comparison solution as the baseline failure probability. Subsequently, the RBF-PSO method was executed independently 50 times to obtain the average failure probability. As shown in Table 2, compared to other reliability analysis results in the literature, the proposed RBF-PSO method demonstrates significant optimization in accuracy and efficiency, with $\Delta \hat{P}_f$ of merely 0.082% and an N_{call} of only 18.14. As shown in Fig. 4, when considering two cases of initial sample points ($N_{\text{doe}}=12,16$), it was observed that as the quantity of sample points increases, the failure probability $\hat{P}_f$ stabilizes and approaches the comparison solution.

To analyze the accuracy and efficiency of the RBF-PSO-IS metamodel, we initially conducted 50 independent experiments using the proposed RBF-PSO-IS method, obtaining the average failure probability, as shown in Table 3. Due to the insufficient accuracy of the LHS in the RBF-PSO stage, adopting the RBF-PSO-IS method based on RBF-PSO brings the results closer to the analytical solution. As shown in Fig. 4, when examining two cases of initial sample points ($N_{\text{doe}}=12,16$), it was observed that as the quantity of sample points increases, the failure probability $\hat{P}_f$ tends to stabilize and approaches the analytical solution.

Table 2
Results of different methods (stage RBF-PSO).

Method	N_{call}	$\hat{P}_f (\times 10^{-5})$	$\Delta \hat{P}_f (\%)$	$\delta_{\text{MCS}} (\%)$	Reference
MCS*	5×10^7	2.850	1.05	9.40	[28–30]
AK-IS	26	2.860	0.7	2.39	[28]
AK-SS	19	2.850	0	9.76	[29]
FORM + IS	$19 + 10^4$	3.9961	0.715	3.87	[30]
LHS*	5×10^7	2.816	–	2.66	–
RBF-PSO	Case 1	12 + 5 = 17	2.814	0.071	2.67
	Case 2	16 + 5 = 21	2.812	0.142	2.67
	Average50	12 + 6.14 = 18.14	2.81832	0.082	2.66
RBF-GA	12 + 5 = 17	2.814	0.071	2.67	–

Note: $\Delta \hat{P}_f = |\hat{P}_f - \hat{P}_{f\text{MCS}}| / \hat{P}_{f\text{MCS}}$, $\Delta \hat{P}_f = |\hat{P}_f - \hat{P}_{f\text{LHS}}| / \hat{P}_{f\text{LHS}}$, where $\hat{P}_{f\text{MCS}}$ and $\hat{P}_{f\text{LHS}}$ represent the failure probabilities determined by the MCS and LHS methods, respectively. These probabilities serve as reference values (baseline failure probabilities).

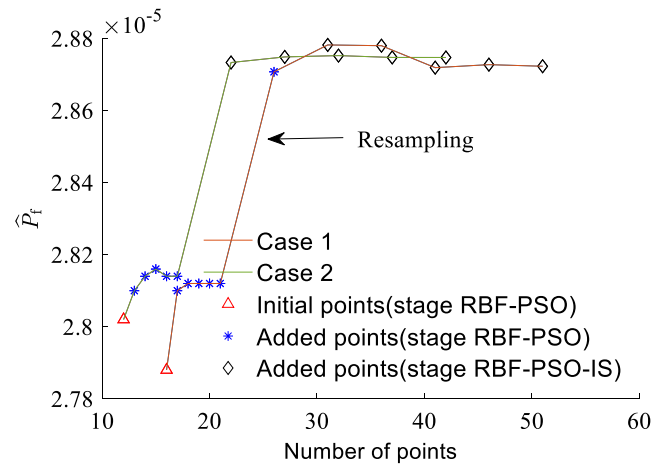


Fig. 4. Convergence for estimation of failure probability $\hat{P}_f$.

Table 3
Results of different methods (stage RBF-PSO-IS).

Method	N_{call}	$\hat{P}_f (\times 10^{-5})$	$\Delta \hat{P}_{f2} (\%)^a$
Analytical Solution*	–	2.87454	–
RBF-PSO-IS	Case 1	12 + 5 + 25 = 42	2.87472
	Case 2	16 + 5 + 30 = 51	2.87229
	Average50	12 + 6.14 + 25.3 = 43.44	2.87424

* Analytical Solution: Calculated using the built-in function “NIntegrate” in Mathematica, and the error is less than $1e-10$, the result can be considered as the analytical solution.

^a $\Delta \hat{P}_{f2} = |\hat{P}_f - \hat{P}_{f\text{Analytical Solution}}| / \hat{P}_{f\text{Analytical Solution}}$, where $\hat{P}_{f\text{Analytical Solution}}$ represents the failure probability under the analytical solution, which is used as a reference value (baseline failure probability).

Fig. 5 illustrates the distribution of initial sample points (black triangles) in Case 1. First, we used the k-means algorithm to divide the sample points, then selected the two sample points with the lowest and highest dispersion in each group, ultimately obtaining all initial sample points. Fig. 6 displays the subsequent sample points (yellow stars) added in the RBF-PSO stage for Case 1. It can be observed that the added subsequent sample points are more discretely distributed around the LSF, which is due to the distance constraint $d_{it} \geq D$. Fig. 7 presents the

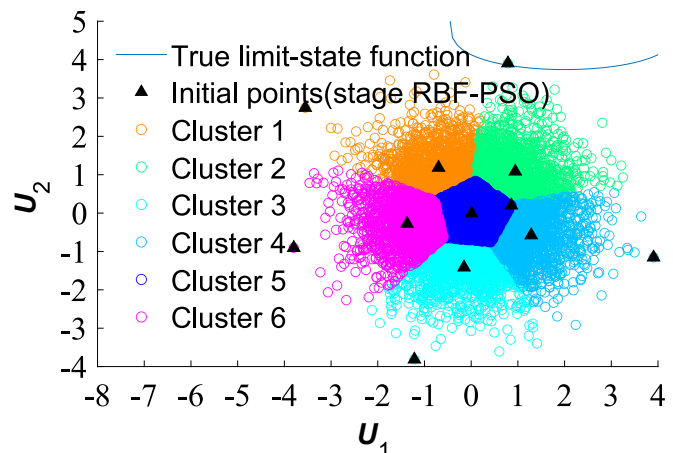


Fig. 5. Case 1 initial points (stage RBF-PSO).

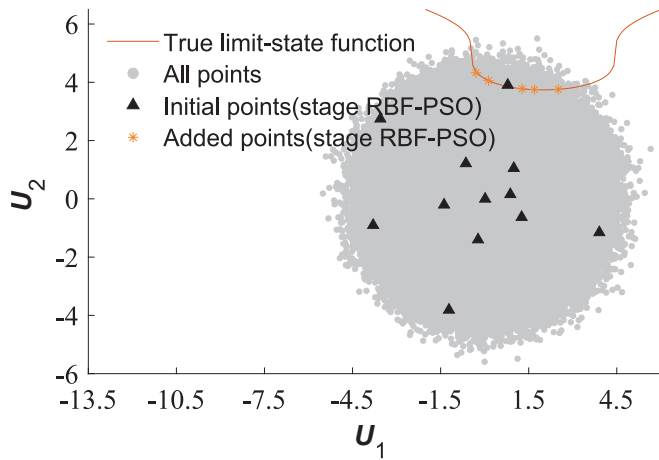


Fig. 6. Case 1 added points process (stage RBF-PSO).

resampled sample points (solid gray circles) and the added subsequent sample points (blue stars) in the RBF-PSO-IS stage for Case 1. The research findings indicate that the resampled sample points are primarily concentrated near the LSF, representing the high-value sample.

Lastly, to verify the RBF-PSO-IS metamodel's accuracy and the applicability of different reliability methods within the RBF-PSO-IS metamodel, we compared the failure probabilities of the LSF and RBF-PSO-IS metamodels under various reliability methods. As shown in Table 4, the relative difference between the failure probabilities $\hat{P}_f$ of RBF-PSO-IS and LSF under different reliability methods is slight,

indicating that the constructed RBF-PSO-IS metamodel effectively represents the LSF. Moreover, the RBF-PSO-IS metamodel using the IS method demonstrates a minor variance, with a $\Delta\hat{P}_{f2}$ of merely 0.001%, suggesting that the constructed metamodel exhibits greater accuracy than those proposed in other studies [10,24,28,30,31].

5. VIV reliability analysis

5.1. A suspension bridge with steel–concrete composite beam

The second application is the Qiupu River Bridge in China. As shown in Fig. 8 and Fig. 9, the Qiupu River Bridge is a large-span suspension bridge, featuring a double simple-supported design. The main span measures 270 m, with the side spans reaching 98 m. The bridge deck comprises a steel–concrete composite girder, measuring 29.5 m in width and 4.34 m in height.

The main girder features a width-to-height ratio of 10 and an opening ratio of 84.7%. Li et al. [32] discovered that the main beam with a height-to-span ratio of 10 and an opening ratio of 90% (similar to the main girder cross-section in this study) displayed a pronounced vertical vortex excitation resonance phenomenon during wind tunnel tests. In conclusion, greater attention should be paid to the vertical VIV phenomenon in steel–concrete composite beams discussed in this paper.

The cross-sectional [33] VIV system is simplified to a two-dimensional vibrating (vertical and torsional) spring-mass-damping system, as shown in Fig. 10. This simplified model is employed for VIV simulation. The model is fixed for calculation; ensure the model cross-section is fully subjected to flow-around. Subsequently, the model is suddenly released to enable motion under aerodynamic forces. This

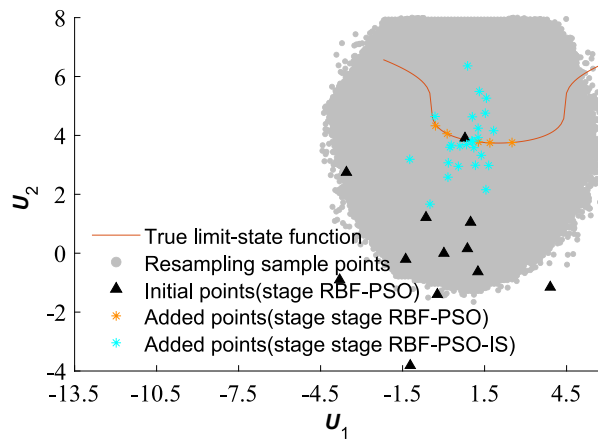


Fig. 7. Case 1 added points process (stage RBF-PSO-IS).

Table 4
Results of different methods.

Method		N_{call}		$\hat{P}_f (\times 10^{-5})$	$\Delta\hat{P}_{f2}(\%)$	Standard deviation ($\times 10^{-7}$)
Analytical Solution *		–		2.87454	–	–
LSF	DS	Average 50	3×10^8	2.88231	0.270	2.81539
RBF- PSO-IS			$12 + 6.14 + 25.3 = 43.44$	2.88222	0.267	2.82076
LSF	LHS	Average 50	3×10^8	2.87963	0.177	2.32167
RBF- PSO-IS			$12 + 6.14 + 25.3 = 43.44$	2.87955	0.174	2.04445
LSF	IS	Average 50	3×10^8	2.87448	0.002	0.03483
RBF- PSO-IS			$12 + 6.14 + 25.3 = 43.44$	2.87452	0.001	0.03958
LSF	JC	–		4.20498	46.284	–
RBF- PSO-IS		Average 50	$12 + 6.14 + 25.3 = 43.44$	4.20463	46.271	–
LSF	Breitung	–		3.06076	6.478	–
RBF- PSO-IS		Average 50	$12 + 6.14 + 25.3 = 43.44$	3.05560	6.299	–
LSF	IS + Breitung	Average 50	1×10^4	2.87723	0.094	0.6196
RBF- PSO-IS			$12 + 6.14 + 25.3 = 43.44$	2.84956	2.653	1.02795

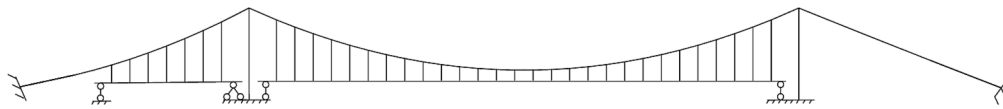


Fig. 8. Bridge elevation layout.

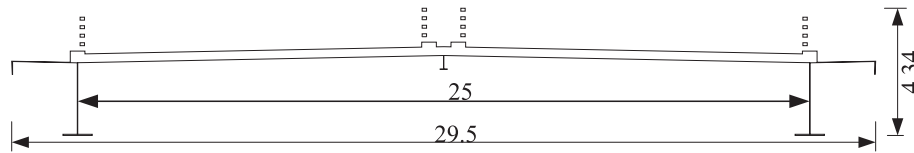


Fig. 9. Qiupu River Bridge cross-section/m.

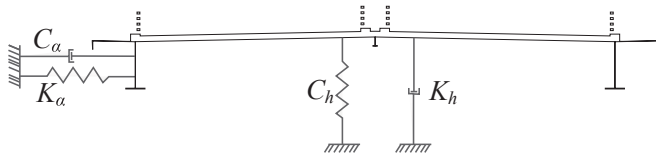


Fig. 10. Two-dimensional vibration simplified model.

process leads to fluid–structure interaction.

The Computational Fluid Dynamics (CFD) software used in this study shows a high degree of confidence when compared with experimental results. In this study, the interleaved method [34,35] is utilized for fluid–structure interaction, which adds two steps of prediction and correction of flow field forces to the traditional method. This approach eliminates iterations and is flexible enough to combine various fluid and solid solvers. The structure is solved using the finite element method, while the fluid is solved using the finite volume method. The SSTk- ω (k represents the turbulent kinetic energy and ω represents the characteristic dissipation velocity) turbulence model is employed for the numerical simulation, with fluid–solid coupling used for the solution to identify the structural VIV response. The outer boundary of the computational region is rectangular (Fig. 11). A dense grid is applied around the delineated main beam section and near the wake flow. A discrete grid around the computational domain, with a reasonable transition in the middle region, is used. The mesh employs an unstructured triangular grid to meet the grid size $Y^+ < 7$ requirement near the wall.

5.2. Parameters for the numerical wind tunnel test

The wind speed interval for calculation was set from 0.5 to 4 m/s, corresponding to the actual bridge wind speeds of 3.536–28.284 m/s. Vertical VIV simulations were conducted at calculated wind speeds of 0.5, 1.0, 2, 2.5, 2.75, 3, and 4 m/s. To account for structural uncertainties [36], the parameters were considered, and their values can be found in Table 5. The Reynolds number at 3 m/s calculated wind speed is $Re = 119793$, and 99.8% of boundary points have Y^+ value less than 7. (Fig. 13), meeting the mesh size requirements near the wall. The total number of mesh triangles is 51,279, and the number of nodes is 26,889. Four multiple mesh layers are employed, with a minimum mesh size of 0.0006 m. Boundary conditions are illustrated in Fig. 12.

5.3. Uncertainty analysis of VIV

The numerical results of the CFD simulation are summarized and transformed for the actual bridge. For bridges with different damping, the vertical displacement changes with the wind attack angle and wind

Table 5
Calculation parameters.

Parameters	Real bridge value	Scaling Ratio	Model Value
Width B /m	29.50	1/50	0.59
Height D_h /m	2.950	1/50	0.059
Mass per Unit Length m /(kg·m ⁻¹)	35,366	1/50 ²	14.146
Bending frequency f_h /Hz	0.293	50/ $\sqrt{50}$	1.831
Bending damping ratio ξ_h /%	0.01	0.5	0.005

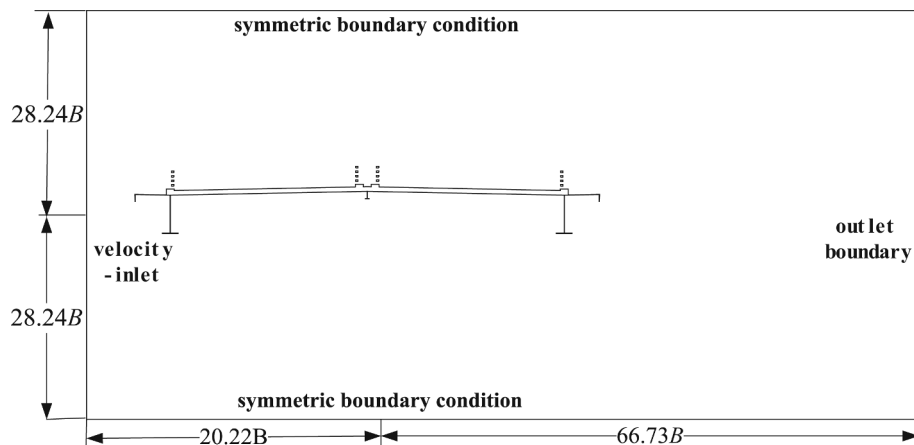


Fig. 11. Calculation area.

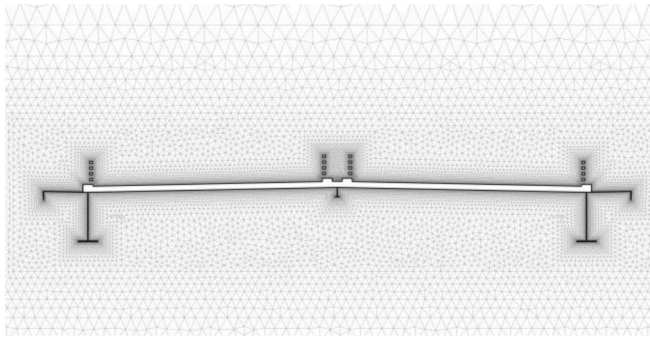


Fig. 12. Flow field grid.

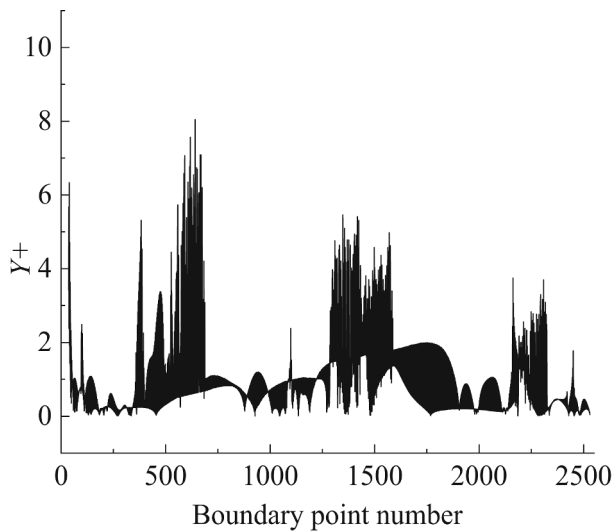


Fig. 13. Distribution.

speed, as shown in Fig. 14. Each surface is based on triangulation-based cubic interpolation to obtain the fitting surface.

Bridge VIV usually occurs in a specific wind direction [37–39]. Fig. 15 shows a schematic of the wind direction in which VIV occurs. The

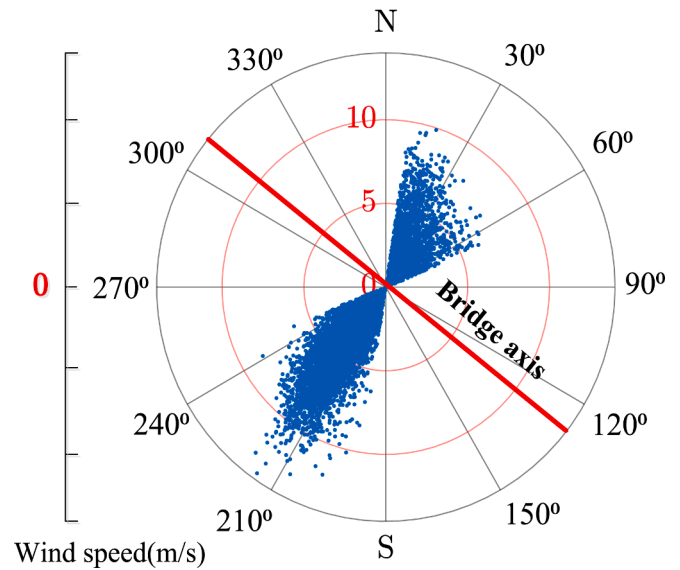


Fig. 15. Wind direction for VIV occurrence.

wind direction of a VIV event at a bridge (blue dots) corresponds to a direction almost perpendicular to the longitudinal bridge axis (red line).

In this study, VIV analysis considers uncertainties in wind characteristics, structural properties, and wind-bridge interactions. By aggregating hourly wind speed and wind attack angle data from the Wangjiang area in China between 1979 and 2020, the corresponding nonparametric probability densities were obtained, as shown in Fig. 16. In the reliability analysis of VIV, uncertainties of wind speed, wind direction, and wind attack angle were introduced. The uncertainties in structural dynamic response due to changes in structural properties warrant particular attention; thus, this study also considered the uncertainties in structural damping and displacement limits. The parameter distribution types and their standard deviation are summarized in Table 6.

5.4. Probability of damage for maximum VIV amplitude

In this sub-section, the damage probability of the VIV maximum

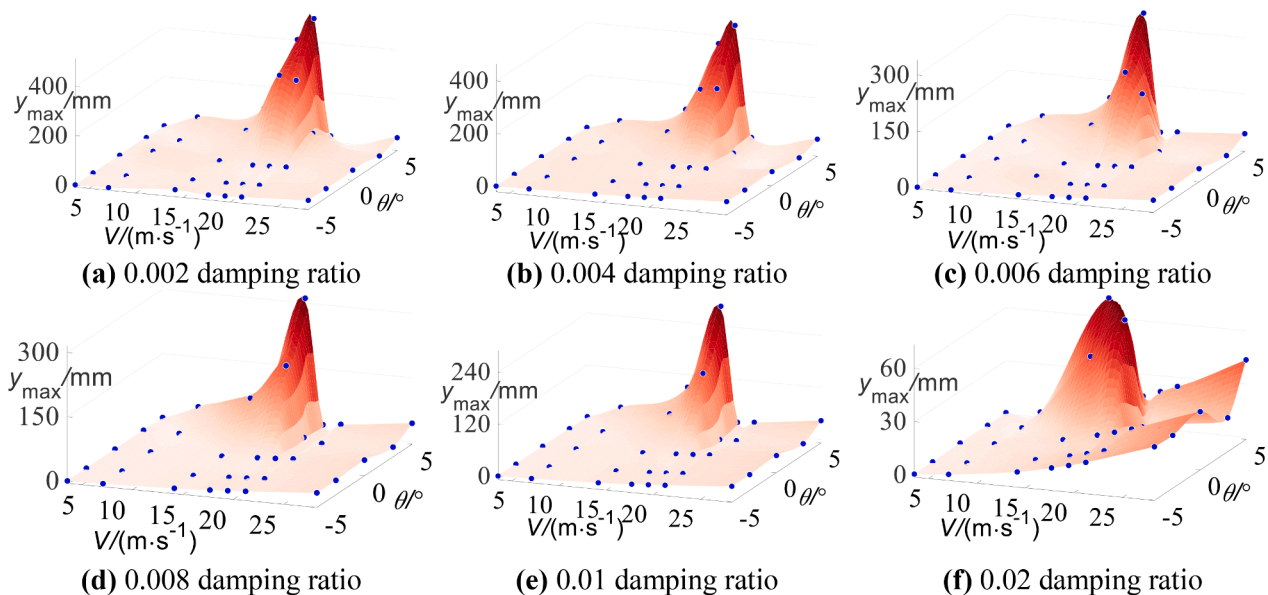


Fig. 14. Variation of vertical displacement with wind angle and speed.

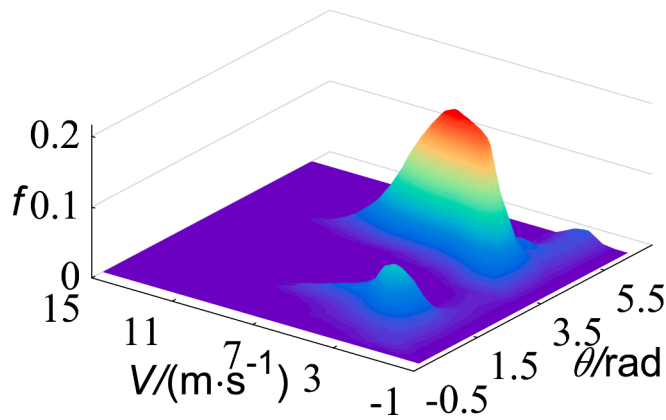


Fig. 16. Probability density function of wind speed and wind attack angle.

Table 6
Distribution of basic variables.

Item	Description	Distribution	Mean value	standard deviation	Reference
$\theta/^\circ$	wind attack angle	Normal	0	1	[3]
ξ_h	Damping ratio	Lognormal	0.005	0.4×0.005	[40]
$[y_a]/\text{mm}$	displacement limit	Lognormal	362.415	0.15×362.415	assumed
α/rad	wind direction	MCMC sampling	–	–	[41]
$V/(\text{m}\cdot\text{s}^{-1})$	wind speed	–	–	–	–

amplitude is considered for the case where the damage criterion has ambiguity $[y_a]$ and uncertainty in the wind attack angle θ and bridge damping ξ_h . The maximum vertical amplitude A of the real bridge for all ξ_h in the numerical simulations at different θ are aggregated and fitted to the equations. The coefficient of determination is $R^2 = 0.99$, and the specific equations are shown in Eq. (13).

$$\begin{cases} \ln(y_{\max}) = (a + b\xi_h + c e^{(\xi_h/11.018)} + d\xi_h/\ln(\xi_h) + p\theta + q\theta^2 + g\theta^3 + w e^{(-\theta)}) \\ a = 1382974.4, b = 125156.57, c = -1382970.1, d = -1517.989 \\ p = 0.425, q = 0.116, g = -0.022, w = -0.025 \end{cases} \quad (13)$$

where, the given values $\xi_h = 0.005$ and $W = 0.005$ are substituted into Eq. (1) to obtain the mean value of $[y_a]$. Then, based on Eq. (2) and (14), we calculate the LSF (see Fig. 17).

This study employs the LHS method to generate 1×10^7 sample points and establish a comparative solution as a baseline failure probability. Subsequently, the proposed RBF-PSO method is executed independently 50 times to obtain the average failure probability, as

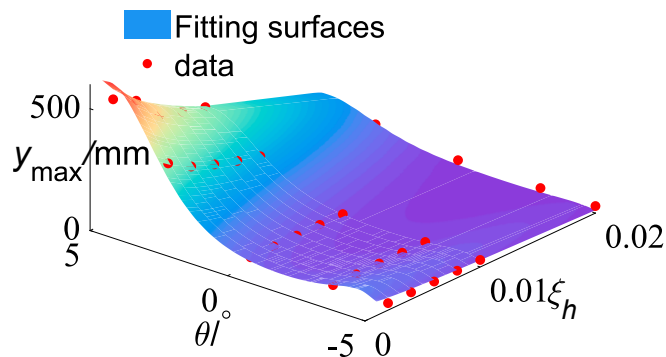


Fig. 17. Probability density function of wind speed and wind attack angle.

illustrated in Table 7. Compared to the comparative solution, $\Delta\hat{P}_f(\%)$ under the RBF-PSO method is only 0.832%, demonstrating sufficient accuracy to meet engineering requirements. As shown in Fig. 18, considering two cases with initial sample points ($N_{\text{doe}}=30, 40$), it is found that as the number of sample points increases, the failure probability $\hat{P}_f$ stabilizes and approaches the comparative solution.

Initially, 50 independent experiments were conducted using the proposed RBF-PSO-IS method, obtaining an average failure probability, as displayed in Table 8. Utilizing LHS to generate 5×10^{10} sample points as a comparison solution, according to Eq. (14), the error of the corresponding $\hat{P}_f$ has 99% probability of being less than 0.002%. As depicted in Fig. 18, two cases with initial sample points ($N_{\text{doe}}=30, 40$) were considered. As the number of sample points increased, the failure probability stabilized and approached the comparison solution.

$$V_p = \sqrt{\frac{\Phi(-2\beta)\exp(\beta^2) - \Phi^2(\beta)}{N_1\Phi^2(-\beta)}} \quad (14)$$

where, $\beta = \Phi^{-1}(1 - \hat{P}_f)$, and $V_p = \varepsilon_{is}/\mu_{\alpha/2}$ (ε_{is} is the error of $\hat{P}_f$, $\mu_{\alpha/2}$ is the upper quintile of the standard normal distribution $\alpha/2$).

Figs. 19–21 adopt the same illustration method as Figs. 5–7 but the transition from a two-dimensional to a three-dimensional presentation. Observations reveal that, during the RBF-PSO stage, the added subsequent sample points (red triangles) are discretely distributed near the LSF. In contrast, the resampled sample points (gray circles) are concentrated near the LSF. Whether the data state is two-dimensional or three-dimensional, the RBF-PSO-IS method performs well and effectively screens for high-value sample points.

Lastly, the LSF and the final RBF-PSO-IS model were applied separately to various reliability methods to calculate the failure probability. As shown in Table 9, for different reliability methods, the error in failure probability $\hat{P}_f$ between RBF-PSO-IS and LSF is relatively small, indicating that the constructed RBF-PSO-IS model effectively represents the LSF. Furthermore, the RBF-PSO-IS method incorporating IS exhibits minimal variance, with $\Delta\hat{P}_{f2}$ at merely 0.001%, further confirming the excellent performance of the constructed metamodel.

5.5. Damage probability in VIV interval

In addition to the ambiguity of failure criteria $[y_a]$ and the uncertainty of wind attack angle and damping ratio, the influence of wind speed and direction on the damage probability of the VIV interval should also be considered. In this case, it is challenging to perform curve fitting using a definite equation, so we choose the interpolation method and use the “intern” function from the Matlab package. Next, obtain an implicit equation $y_{\max}(\theta, \xi_h, \alpha, V)$, and as shown in Fig. 1, VIV will not occur when α is not within the VIV wind direction range. Utilizing Eq. (2) and referencing Table 5, the implicit LSF is obtained. (The value of ξ_h is 0.003, and $W = 2$ is substituted into Eq. (1) to obtain the mean value of $[y_a]$).

To analyze the accuracy and efficiency of the adaptive sampling strategy proposed in this study and the influence of the number of initial sample points in stage PSO-RBF, 2.5×10^7 random sample execution

Table 7
Results of different methods (stage RBF-PSO).

Method	N_{call}	$\hat{P}_f (\times 10^{-4})$	$\Delta\hat{P}_f(\%)$	$\delta_{\text{MCS}}(\%)$
LHS	1×10^7	2.402	–	2.66
RBF- PSO	Case 1	30 + 18 = 48	2.384	0.749
	Case 2	40 + 16 = 21	2.394	0.333
RBF- GA	Median 50	30 + 19.5 = 49.5	2.422	0.832
	30 + 18 = 48		2.384	0.749

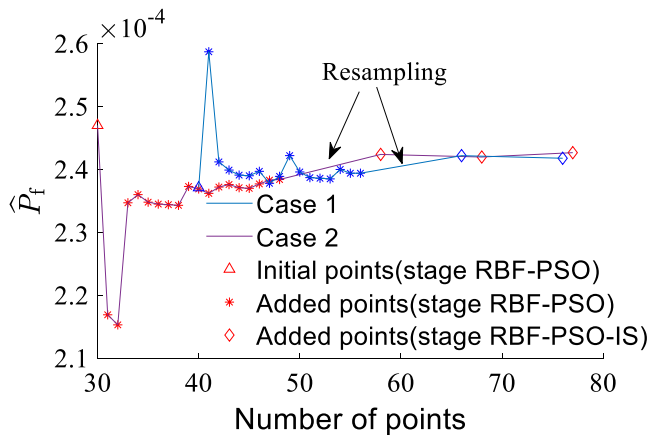


Fig. 18. Convergence for estimation of failure probability.

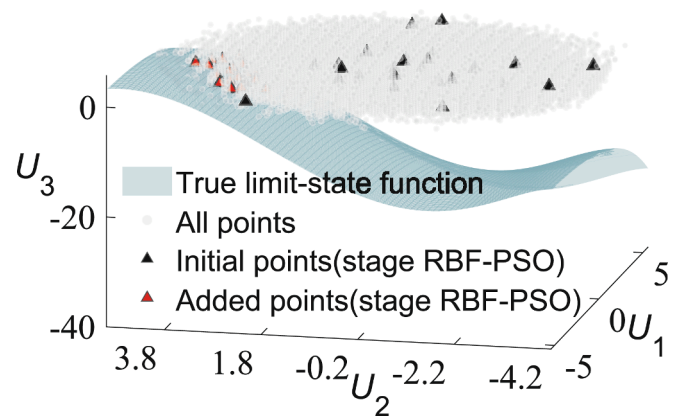


Fig. 20. Case 1 added points process (stage RBF-PSO).

Table 8

Results of different methods (stage RBF-PSO-IS).

Method	N_{call}	$\hat{P}_f (\times 10^{-4})$	$\Delta \hat{P}_{f2} (\%)^a$
LSF (Contrast Solution)*	5×10^{10}	2.42228	–
RBF- PSO-IS	Case 1	30 + 18 + 29 = 77	2.42669
	Case 2	40 + 16 + 20 = 76	2.41778
	Median	30 + 19.5 + 28.62 = 78.12	2.42351
	50		0.051

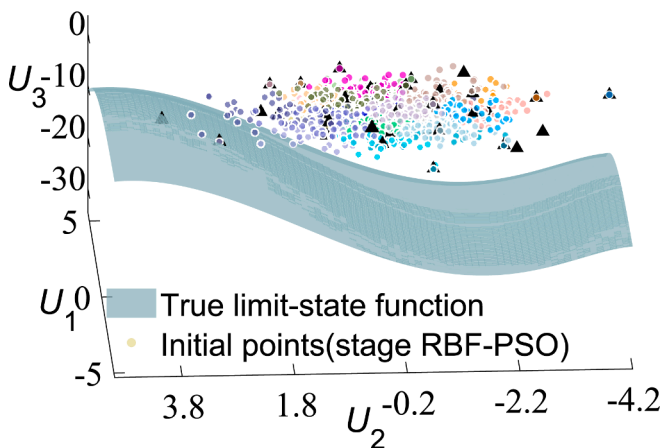


Fig. 19. Case1 initial points (stage RBF-PSO).

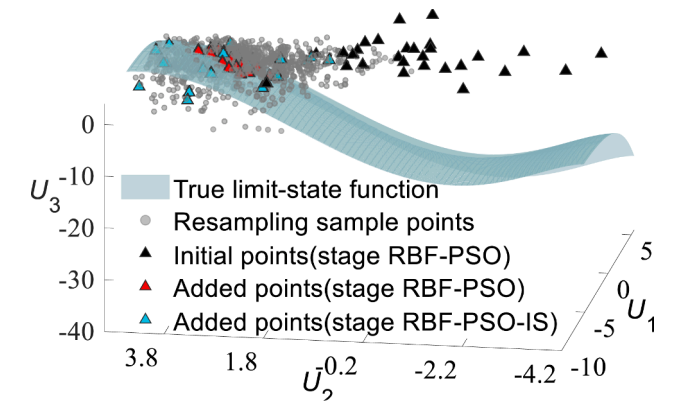


Fig. 21. Case 1 added points process (stage RBF-PSO-IS). note: U_1 , U_2 , and U_3 represent the distribution conditions of θ , ξ_h , and $[y_a]$, respectively, in the standard normal space.

Table 9

Results of different methods.

Method	N_{call}	$\hat{P}_f (\times 10^{-4})$	$\Delta \hat{P}_{f2} (\%)$
LSF (Contrast Solution)	–	5×10^{10}	2.42228
LSF	DS	Median	1×10^8
RBF- PSO-IS	50	30 + 19.5 + 28.62 = 78.12	2.41901
LSF	LHS	Median	1×10^8
RBF- PSO-IS	50	30 + 19.5 + 28.62 = 78.12	2.42284
LSF	IS	Median	1×10^8
RBF- PSO-IS	50	30 + 19.5 + 28.62 = 78.12	2.42523
LSF	JC	–	1×10^8
RBF- PSO-IS	50	30 + 19.5 + 28.62 = 78.12	2.42975
LSF			2.42226
RBF- PSO-IS			2.42274
LSF			2.93554
RBF- PSO-IS			2.93009
			21.189
			20.964

Table 10

Comparison of calculation results of different methods (stage RBF-PSO).

Method	N_{call}	$\hat{P}_f (\times 10^{-5})$	$\Delta \hat{P}_f (\%)$	$\delta_{\text{MCS}} (\%)$
LHS	2.5×10^7	1.65	–	4.923
RBF- PSO	200 + 64 = 264	1.675	1.515	4.886
RBF- GA	200 + 64 = 264	1.675	1.515	4.886

results generated by LHS are employed as a comparative solution. Simultaneously, considering the initial sample points $N_{\text{doe}} = 200$, RBF is enhanced using the PSO and GA methods. The results are presented in Table 10. Fig. 22 illustrates the outcomes of the PSO-RBF method when the IS method is applied. However, applying the IS method in the PSO-RBF-IS stage is challenging due to the excessively high nonlinearity of the constructed metamodel. Consequently, only the PSO-RBF method is utilized to calculate the failure probability of VIV, which is adequate to meet the engineering requirements.

6. Concluding remarks

A novel scheme, RBF-PSO-IS, is presented for investigating the VIV failure probability of steel–concrete composite beam bridges, based on RBF neural network, adaptive sampling strategy, and optimization using

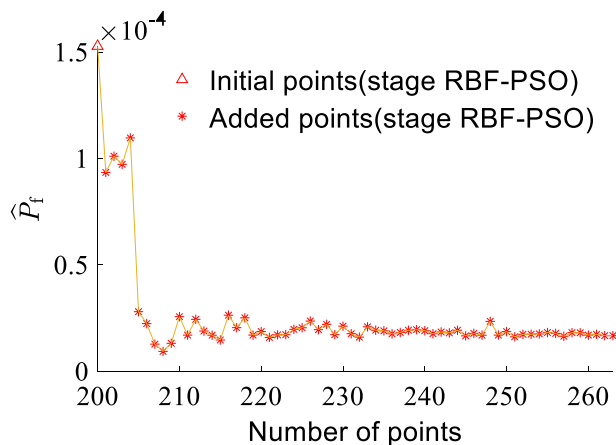


Fig. 22. Convergence for estimation of failure probability $\hat{P}_f$.

PSO and IS. This paper employs an RBF neural network to replace the complex and time-consuming CFD flow simulation process. It introduces an adaptive random sampling method and evolutionary algorithm to enhance the efficiency of neural network modeling, change the sampling center of gravity through significant sampling, and reconstruct the metamodel. Ultimately, it combines multiple reliability algorithms to calculate VIV's maximum amplitude and the VIV interval's failure probability.

The VIV reliability analysis method proposed in this paper transforms the problem of fitting the VIV instability LSF in the full domain of the variable design space using RBF neural networks into the problem of fitting the LSF. This reduces computational complexity and cost while ensuring accuracy, resulting in higher accuracy than previous studies.

The IS method is employed to reconstruct a more accurate metamodel, effectively improving the reliability model sampling results that are difficult to approach the LSF. The failure probability is very close to the analytical solution.

It is worth noting that this research method does not make specific assumptions about the nature of structural LSF and random variables. Therefore, by adjusting the numerical model of the method, this approach can be applied to other reliability problems in engineering. However, as the number of dimensions increases, finding a sampling distribution that covers high-probability areas and avoids low-probability areas becomes difficult. Thus, for high-dimensional problems, the RBF-PSO-IS method may be less suitable, while the RBF-PSO method remains applicable, although its accuracy might only meet engineering requirements.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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